Comparing groups

REFERENCE SHEET

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```r
library(tidyverse)
library(flextable)
library(ggridges)
```

```r
wq <- read_csv("https://raw.githubusercontent.com/ENVS-193DD/datasets/refs/h
↪ eads/main/water-quality/wq_clean.csv") |>
  mutate(site_name = fct_relevel(site_name,
                                 "Phelps Bridge", "East Channel", "Venoco
                                 ↪ Bridge"))
```

```r
# colors to use in all figures
site_cols <- c(
  "East Channel" = "orange4",
  "Phelps Bridge" = "tomato3",
  "Venoco Bridge" = "turquoise4"
)

# setting common theme elements
theme_set(
  theme_classic() +
    theme(text = element_text(size = 12),
          plot.title.position = "plot",
          legend.position = "none")
)
```

1. Math

Mean

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$\bar{y}$ is the sample mean, which is calculated as the sum of all observations y_i divided by the number of observations (n).

Median

The median is determined by finding the middle value. If n is odd, the median is the middle value in the dataset. If n is even, the median is the mean of the two middle values.

Standard deviation

$$s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$$

s is the sample standard deviation, which is calculated as the square root of the sum of the squared difference between an observation y_i and the sample mean $\bar{y}$ divided by the number of observations n minus 1.

Variance

$$s^2 = \frac{\sum(y_i - \bar{y})^2}{n-1}$$

Variance is calculated as the standard deviation squared. Note that the units of variance are the units squared (e.g. g^2, m^2, ppt^2).

2. Water quality

a. Calculating central tendency and spread

```
# creating new object from water quality
salinity_summary <- wq |>
  # group by site
  group_by(site_name) |>
  # calculate mean, median, standard deviation, variance, number of obs
  summarize(mean_sal = mean(mean_salinity_ppt),
            median_sal = median(mean_salinity_ppt),
            sd_sal = sd(mean_salinity_ppt),
            var_sal = var(mean_salinity_ppt),
            n = length(mean_salinity_ppt)) |>
  # ungroup data frame
  ungroup()

# use flextable to create table
flextable(salinity_summary) |>
  # change column names in table to be readable
  set_header_labels(
    site_name = "Site",
    mean_sal = "Mean",
    median_sal = "Median",
    sd_sal = "Standard deviation",
    var_sal = "Variance",
    n = "Number of surveys"
  ) |>
  # round any columns with numbers to one decimal point
  colformat_double(j = c("mean_sal", "median_sal", "sd_sal", "var_sal"),
                   digits = 1) |>
  # setting column width of table
  width(width = 1,
```

```
        unit = "in")
```

Site	Mean	Median	Standard deviation	Variance	Number of surveys
Phelps Bridge	3.7	2.2	5.3	27.9	234
East Channel	19.0	19.7	16.2	262.3	180
Venoco Bridge	22.7	23.4	18.3	336.0	242

b. Visualizing distributions

Purpose: represent the distribution of variable of interest

Histograms

Components

- x-axis: *continuous* variable of interest
- y-axis: counts
- Column height represents number of observations falling into a given bin range on the x-axis

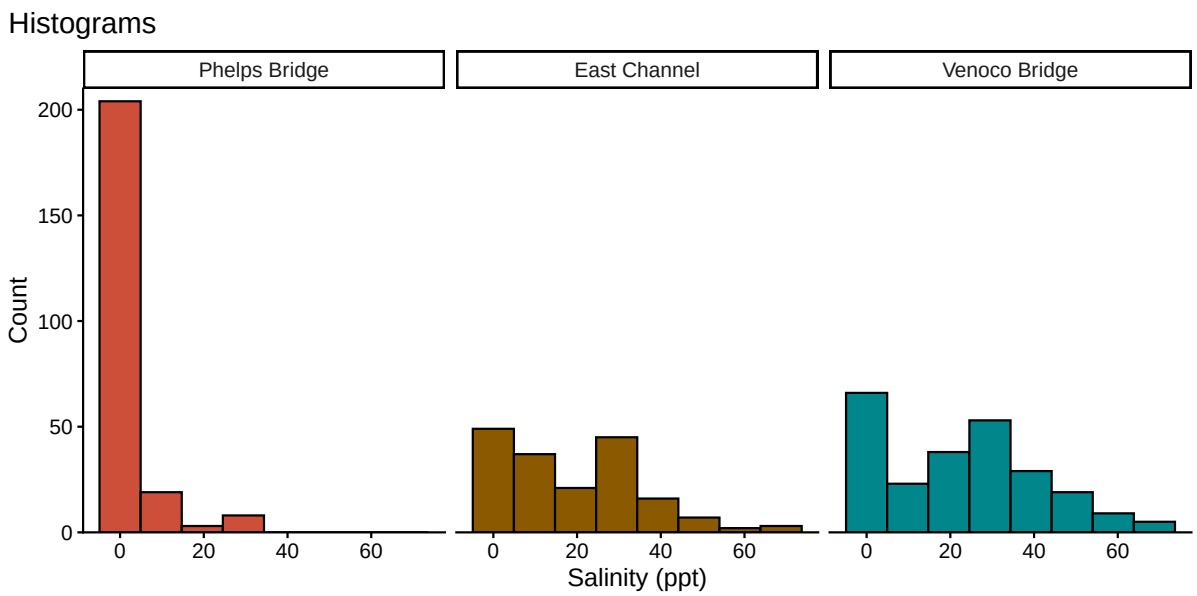
Pay attention to:

- the shape of the distribution
- the center of the distribution
- the peak(s) of the distribution

```
# creating object for histogram
histogram <- ggplot(data = wq,
                    # x-axis: variable of interest
                    # fill geometries by site
                    mapping = aes(x = mean_salinity_ppt,
                                  fill = site_name)) +
  # histogram geometry
  geom_histogram(bins = 8,             # using 8 bins
                 color = "black") +    # coloring outline black
  # creating different panels for each site
  facet_wrap(~ site_name) +
  # getting rid of space between columns and axis, setting limits
```

```r
  scale_y_continuous(expand = c(0, 0),
                     limits = c(0, 210)) +
  # filling columns
  scale_fill_manual(values = site_cols) +
  # relabelling axes, adding title
  labs(x = "Salinity (ppt)",
       y = "Count",
       title = "Histograms")

# displaying histogram
histogram
```



Density plots

Components:

- x-axis: continuous variable of interest
- y-axis: probability density
- smoothed function representing the shape of the distribution
- note: area under that smoothed function = 1

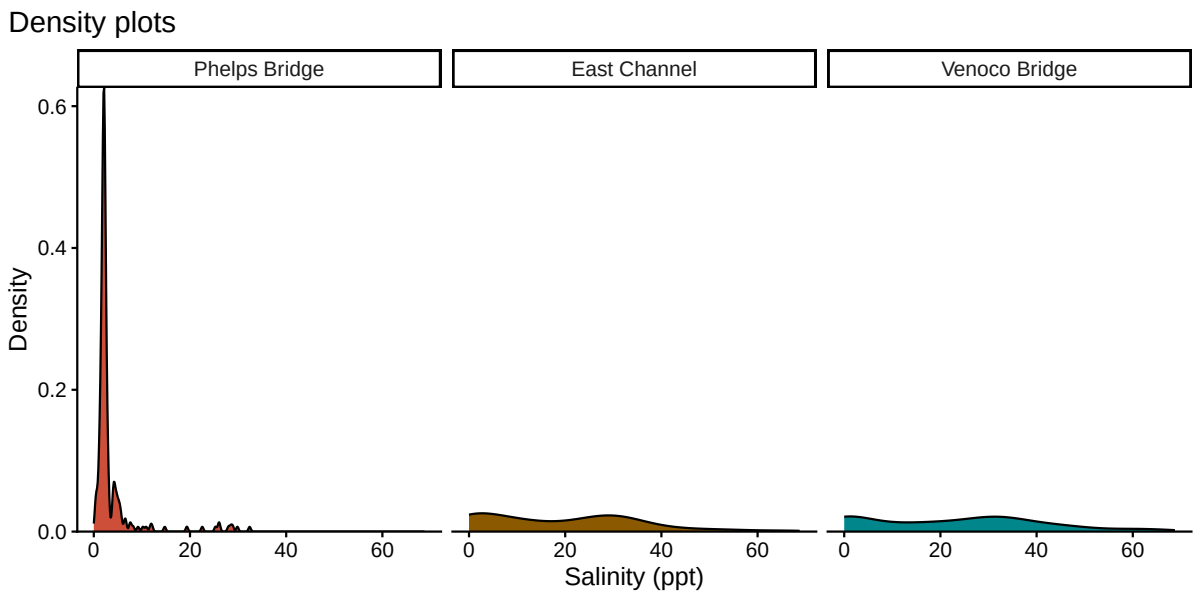
Pay attention to:

- the shape of the **distribution**
- the center of the distribution

- the peak(s) of the distribution

```
density <- ggplot(data = wq,
                  mapping = aes(x = mean_salinity_ppt,
                                fill = site_name)) +
  geom_density() +
  facet_wrap(~ site_name) +
  scale_y_continuous(expand = c(0, 0)) +
  scale_fill_manual(values = site_cols) +
  labs(x = "Salinity (ppt)",
       y = "Density",
       title = "Density plots")

density
```



Interpretation:

Most salinity measurements from Phelps Bridge are low. Most salinity measurements from East Channel and Venoco Bridge are below 40 ppt.

Boxplots

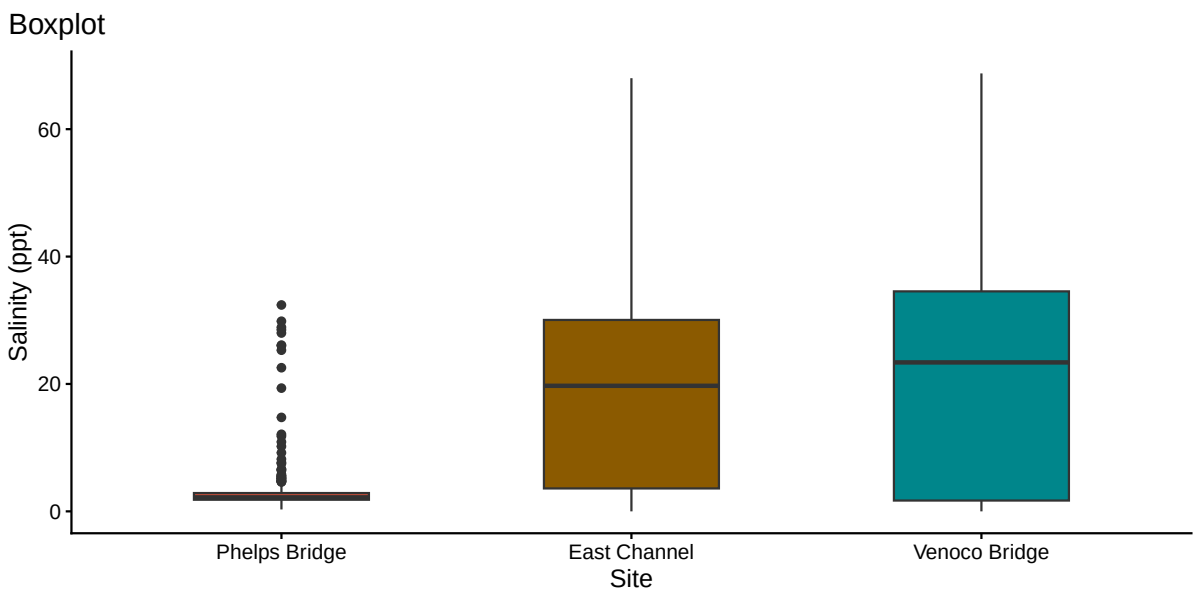
Components:

- x-axis: categories (levels in a factor)
- y-axis: variable of interest

- middle line: median
- bottom of box: first quartile (25th percentile)
- top of box: third quartile (75th percentile)
- whiskers: 1.5 × interquartile range
- points: observation values outside 1.5 × interquartile range

```
boxplot <- ggplot(data = wq,
                  mapping = aes(x = site_name,
                                y = mean_salinity_ppt,
                                fill = site_name)) +
  geom_boxplot(width = 0.5) +
  scale_fill_manual(values = site_cols) +
  labs(x = "Site",
       y = "Salinity (ppt)",
       title = "Boxplot")

boxplot
```



Violin charts

Note: violin charts are density plots that are mirrored.

Components:

- x-axis: categories (levels in a factor)

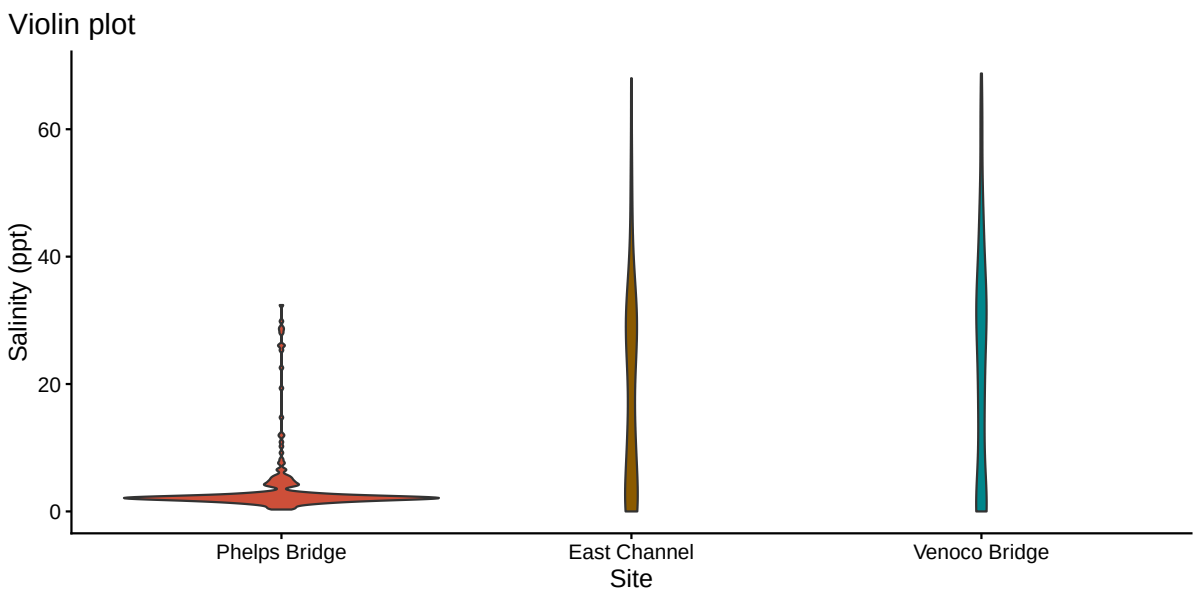
- y-axis: variable of interest
- geometry: jittered points

Pay attention to:

- wide or narrow parts of the violin
- position of wide or narrow parts along y-axis

```
violin <- ggplot(data = wq,
                 mapping = aes(x = site_name,
                               y = mean_salinity_ppt,
                               fill = site_name)) +
  geom_violin() +
  scale_fill_manual(values = site_cols) +
  labs(x = "Site",
       y = "Salinity (ppt)",
       title = "Violin plot")

violin
```



Strip charts

Components:

- x-axis: categories (levels in a factor)

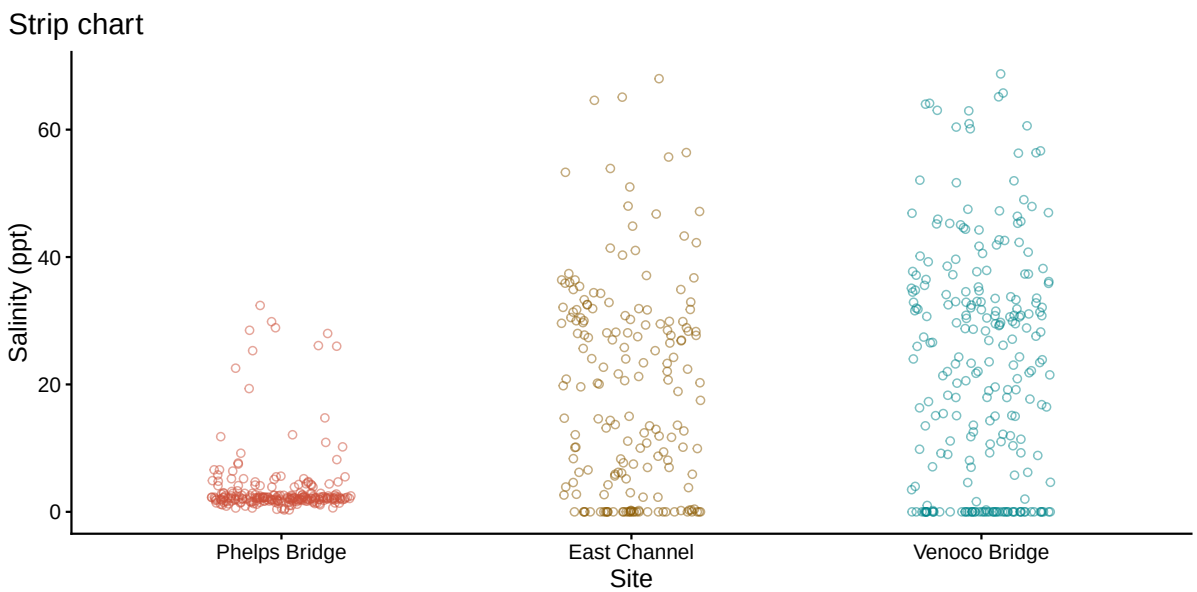
- y-axis: variable of interest
- geometry: jittered points

Pay attention to:

- the clusters of points, if any
- position of points along the y-axis

```
jitter <- ggplot(data = wq,
                 mapping = aes(x = site_name,
                               y = mean_salinity_ppt,
                               color = site_name)) +
  geom_jitter(height = 0,
              alpha = 0.5,
              width = 0.2,
              shape = 21) +
  scale_color_manual(values = site_cols) +
  labs(x = "Site",
       y = "Salinity (ppt)",
       title = "Strip chart")

jitter
```

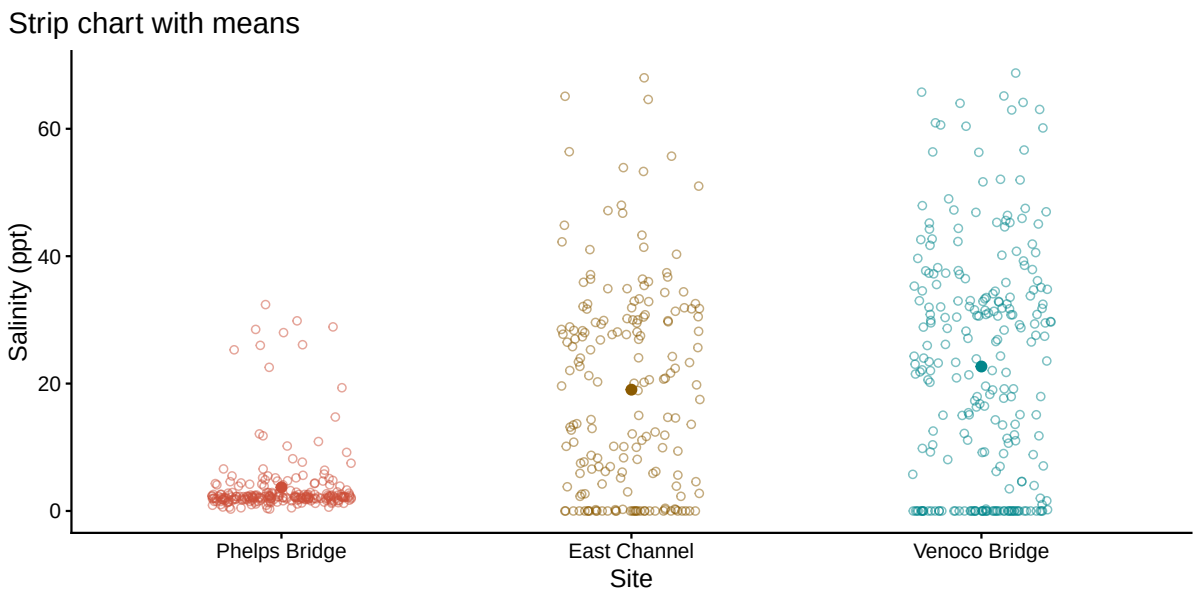


c. Visualizing central tendency and spread

Means

With standard deviation:

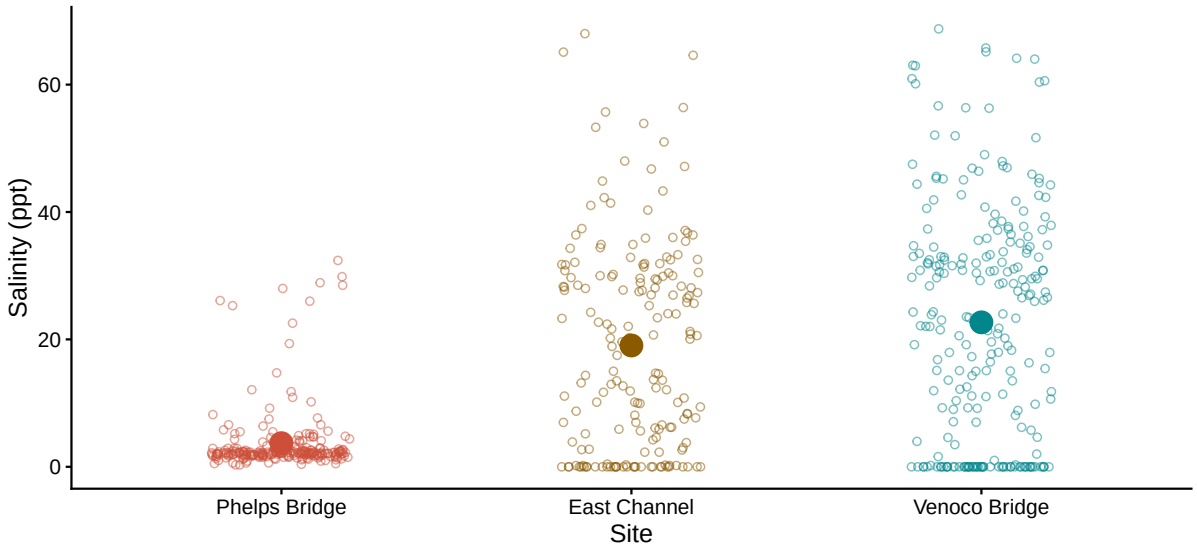
```
jitter +
  geom_point(data = salinity_summary,
             mapping = aes(x = site_name,
                           y = mean_sal,
                           color = site_name),
             size = 2) +
  labs(title = "Strip chart with means")
```



With standard error:

```
jitter +
  stat_summary(geom = "pointrange",
               fun.data = mean_se,
               size = 1,
               linewidth = 1) +
  labs(title = "Strip chart with means and SE")
```

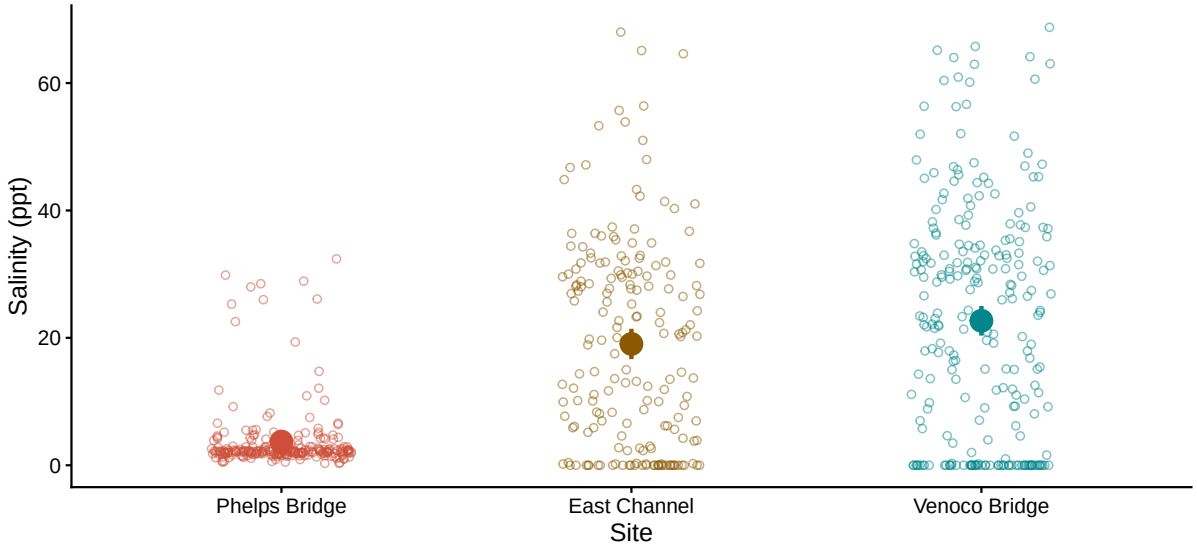
Strip chart with means and SE

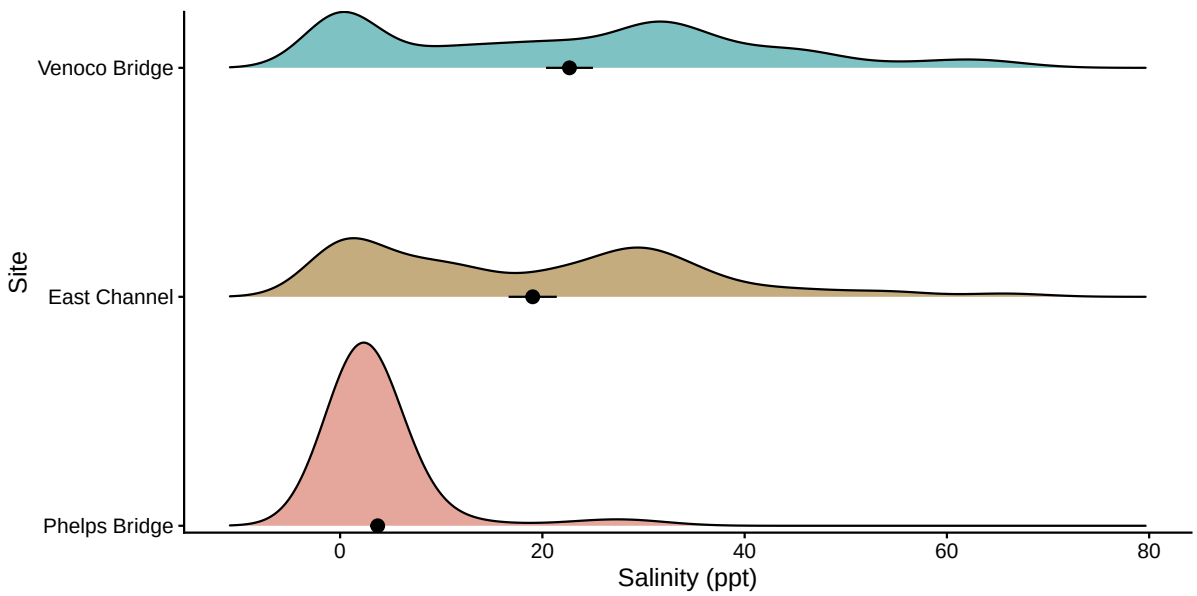


With 95% confidence interval:

```
jitter +
  stat_summary(geom = "pointrange",
               fun.data = mean_cl_normal,
               size = 1,
               linewidth = 1) +
  labs(title = "Strip chart with means and 95% confidence interval")
```

Strip chart with means and 95% confidence interval



d. Exciting

```
ggplot(data = wq,
       mapping = aes(x = mean_salinity_ppt,
                     y = site_name,
                     fill = site_name)) +
  geom_density_ridges(alpha = 0.5,
                      scale = 0.8) +
  stat_summary(geom = "pointrange",
               fun.data = mean_cl_normal,
               size = 0.5) +
  scale_fill_manual(values = site_cols) +
  scale_y_discrete(expand = c(0.01, 0.01)) +
  labs(x = "Salinity (ppt)",
       y = "Site")
```